

TRIAD: Trusted RDF Interoperable Agent Data

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Abstract

Keywords

Knowledge Graph, RDF, FHIR, Agent Memory, Healthcare Data Privacy

1 Introduction

2 Motivation

Hospitals are slow to use agents in clinical work, for four reasons. First, patient data is in many separate systems. Even where hospitals adopt interoperability standards such as FHIR [1], some data is on paper, and other data is behind an interface that nobody outside the building can reach.

Second, an agent cannot keep all of this data in its conversation window. When the context is long, language models miss relevant facts. The agent does not see that a fact changed, and it uses an old fact.

Third, patient data is sensitive. Personally Identifiable Information (PII) is information that can identify a person, such as a name or a health card number. When PII is linked to health data, it becomes Protected Health Information (PHI). An agent must not show PHI to the wrong person.

Fourth, the use of PHI and PII is subject to strict laws, such as HIPAA in the United States, and PIPEDA and PHIPA in Canada. A hospital cannot use a system that does not obey these laws.

An Electronic Health Record (EHR) is designed for people to read. It does not give the agent a small and explicit view of who asserted what and what replaced what. The agent must infer this at runtime. When the inference is wrong, nobody can find the cause. Wrong answers decrease the trust of staff, and hospitals then do not use agents.

2.1 Motivating Example

A patient comes to the emergency room with shortness of breath. At triage, the patient tells the agent that the problem started two days ago. The nurse assesses the patient and records a possible asthma attack. One hour later, the physician examines the patient and revises the assessment to possible heart failure. The physician also tells the agent that the father of the patient died of a heart attack. This fact is not in the EHR yet.

In the evening, the patient asks the agent about the plan for their care. The conversation is now long. The asthma assessment appears many times, and the revision appears once. The agent answers from the old assessment and tells the patient that an inhaler is the plan.

With TRIAD, the agent starts at the patient and finds the current assessment one or two hops away in an RDF [2] knowledge graph. It sees that the physician replaced the assessment of the nurse, and it finds the family history. If its answer is wrong, we can trace the error to a specific relationship. In the session of the patient, the

agent shows only the data that is approved for the patient. Staff who do not care for this patient get no access to their PHI.

3 Team Justification

This project combines the backgrounds of the two team members. Khaled Abuela does his M.Sc. thesis research on memory for language model agents under Professor Shengyao Lu. Asish George has four years of experience with the development of healthcare systems. We selected a problem that we both know from real work, not an artificial one.

References

- [1] HL7 International. 2023. FHIR: Fast Healthcare Interoperability Resources. <https://hl7.org/fhir/>.
- [2] W3C. 2014. RDF 1.1 Concepts and Abstract Syntax. <https://www.w3.org/TR/rdf11-concepts/>.